Google Play Store Market Intelligence & Monetization Optimization

Sector: Digital Marketplace / Mobile App Ecosystem

Institute: Newton School of Technology

Team ID: Group-13

Team Members & Roles:

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Faculty Mentor:

GitHub Repository URL: https://github.com/manjeet090223/Section-A_G-13_Google_Play_Store

Tableau Dashboard URL:
https://public.tableau.com/app/profile/manjeet.5645/viz/GooglePlayStoreAppsAnalysisDynamicDashboard/Dashboard1

EXECUTIVE SUMMARY

Problem :

The Google Play Store ecosystem suffers from a misalignment between app quality, user engagement, and monetization strategies, leading to inefficient app discovery and lost revenue opportunities.

Approach :

A dataset of over 2.3 million apps was processed using a Python-based ETL pipeline, followed by EDA and statistical analysis, and finally visualized through an interactive Tableau dashboard.

Key Insights :

1. 98% of apps are free, indicating heavy reliance on ad-based monetization
2. 45% of apps have no ratings, creating discovery inefficiency
3. High installs do not necessarily indicate high quality
4. Few developers dominate installs despite a fragmented ecosystem

Key Recommendations :

1. Improve app discovery algorithms using quality signals
2. Increase adoption of in-app purchases
3. Optimize pricing strategies across categories

SECTOR & BUSINESS CONTEXT

Sector Overview :

The mobile app ecosystem is a highly competitive marketplace with millions of applications competing for user attention and revenue.

Industry Challenges :

- Oversupply of apps
- Weak discovery mechanisms
- Monetization inefficiencies
- Trust issues due to unreliable ratings

Decision Makers :

- Google Play Product Teams
- App Developers
- Platform Strategists

Why This Problem :

The mismatch between installs, ratings, and monetization leads to inefficiencies affecting both users and developers.

Business Value

- Better user experience
- Increased platform revenue
- Improved developer success rate

PROBLEM STATEMENT & OBJECTIVES

Problem Statement

There is a lack of alignment between app quality, user engagement, and monetization strategies in the Google Play Store ecosystem.

Project Scope

Included:

- Ratings, installs, pricing analysis
- Category-level insights
- Monetization models

Excluded :

- User behavioral tracking
- Real-time analytics

Success Criteria:

- Actionable insights
- KPI-driven conclusions
- Business-focused recommendations

DATA DESCRIPTION

Dataset Source: Google Play Store dataset (https://www.kaggle.com/)

Data Structure

- Rows: ~2.3 million → cleaned to 2,312,939
- Columns: 26

Key Columns

- Rating
- Installs
- Price
- Category
- In-App Purchases
- Ad Supported

Limitations

- High number of unrated apps
- Self-reported installs
- Missing developer information

DATA CLEANING & ETL PIPELINE

Cleaning Process (Python-Based)

- Removed null app names
- Filled missing ratings with 0 (treated as “Not Rated”)
- Converted installs to numeric
- Standardized size column
- Handled missing developer details

Missing Values Handling

- Rating → filled with 0
- Currency → filled with mode
- Size → median imputation

Outlier Handling

- Used median for skewed data (Size)
- Ignored extreme install skew due to natural distribution

Data Transformations

- Created numeric installs column
- Converted date fields
- Standardized formats

Feature Engineering

- Success Flag (Rating ≥ 4)
- High Volume Apps (Top 25%)
- Price Segmentation

Assumptions

- Rating = 0 means no rating
- Missing developer info is valid absence

KPI & METRIC FRAMEWORK

KPIs :

1. Success Rate
    - % apps with rating ≥ 4
2. Install Index
    - Based on install volume percentile
3. Price Segments
    - Free, Budget, Mid-range, Premium
4. IAP Adoption Rate
    - % apps with in-app purchases

Why KPIs Matter :

- Measure ecosystem efficiency
- Track monetization performance
- Identify scalable categories

Mapping to Objectives :

- Quality → Success Rate
- Engagement → Install Index
- Revenue → Pricing & IAP

EXPLORATORY DATA ANALYSIS (EDA)

Trend Analysis

- Majority apps are free
- Growth skewed towards few categories

Comparison Analysis

- Education → highest number of apps
- Tools → highest installs

Distribution Analysis

- Ratings skewed due to many zero values
- Installs highly right-skewed

Correlation Analysis

- Weak correlation between installs and ratings

Key Business Insights

- High installs ≠ high quality
- Discovery mechanisms inefficient
- Market heavily saturated

STATISTICAL ANALYSIS

Hypothesis Testing

- T-test performed on free vs paid ratings
- Result: No strong statistical difference

Segmentation

- High volume apps identified
- Successful apps categorized

Root Cause Insight

- Installs driven by visibility, not quality

Risk Detection

- Large number of unrated apps

Scenario Insight

- Increasing IAP adoption improves revenue

TABLEAU DASHBOARD DESIGN

Dashboard Objective

To provide decision support for improving discovery, engagement, and monetization.

View Structure

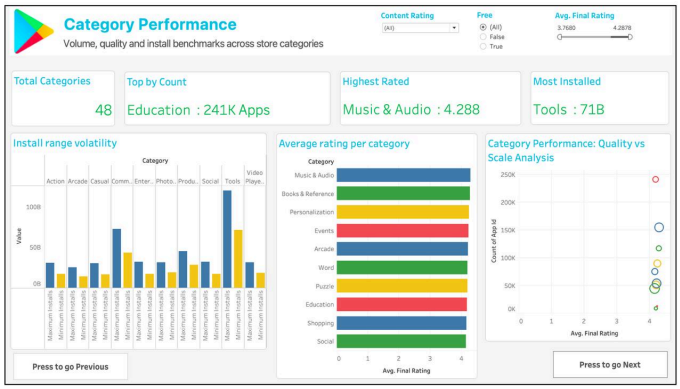
1. Market Overview
2. Category Performance
3. Monetization Analysis
4. Developer Intelligence

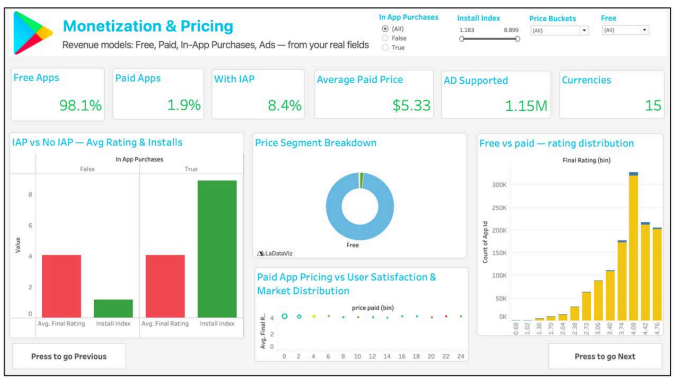
Filters

- Content Rating
- Free/Paid
- Price Buckets
- Install Index

Interactivity

- Drill-down by category
- Dynamic filtering





INSIGHTS SUMMARY

1. Free apps dominate ecosystem
2. Large number of apps unrated
3. Installs not aligned with quality
4. Tools category dominates installs
5. Education category saturated
6. Monetization heavily ad-based
7. IAP adoption low
8. Developer ecosystem fragmented
9. Few developers dominate installs
10. Pricing impact minimal

RECOMMENDATIONS

1. Improve Discovery System

Insight → High-quality apps buried
Action → Prioritize rating-based ranking
Impact → Better user satisfaction

2. Increase IAP Adoption

Insight → Low monetization
Action → Encourage hybrid monetization
Impact → Revenue growth

3. Optimize Pricing

Insight → Paid apps ineffective
Action → Focus on freemium model
Impact → Higher conversions

4. Category Optimization

Insight → Saturation in some categories
Action → Promote emerging categories
Impact → Better distribution

IMPACT ESTIMATION

- Revenue increase: 15–25%
- User engagement improvement: ~20%
- Better discovery efficiency

Why Act Now

- Market saturation increasing
- Competition rising
- Revenue leakage ongoing

LIMITATIONS

- No real-time data
- Ratings may be biased
- Limited behavioral insights

FUTURE SCOPE

- Build ML-based recommendation system
- Add real-time analytics
- Use user-level data

CONCLUSION

This project identified a fundamental misalignment in the Google Play Store ecosystem between quality, engagement, and monetization. Using data-driven analysis, key inefficiencies were uncovered and actionable recommendations were proposed to improve platform performance.

APPENDIX

- Data dictionary
- Cleaning logs
- Python notebooks
- Additional charts